

Who Governs AI While It Is Thinking?

Comparative Runtime Governance and the Economic Wall of Accuracy

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Abstract

Artificial-intelligence governance increasingly addresses model evaluation, routing, output filtering, human oversight, and post-deployment monitoring. Agentic systems, however, can change materially during execution as evidence, model choice, resource use, authority, and proposed actions evolve. This paper proposes comparative runtime governance for consequential AI deployments.

Its central architectural claim is a **Reference-state continuity requirement**: before exceptional risk is known, the system establishes a substantive Reference (baseline) candidate or state, preserves sufficient lineage to relate it to later material changes, and uses that comparative basis when reassessing candidate eligibility before consequential commitment. Higher-fidelity processing remains conditional rather than automatically authoritative. A persistent episode state records decision-relevant candidate changes, resources, evidence provenance, authority, prior interventions, and external effects. A runtime governor separately determines whether another operation may proceed, whether a candidate remains eligible, and which eligible candidate or disposition should determine the outcome; a protected execution gate gives those decisions effect before consequential action.

The paper also develops the **Economic Wall of Accuracy** as a prospective allocation rule under which, after mandatory authority and integrity constraints are satisfied, the next available resource is directed toward the permissible intervention expected to reduce consequential risk most effectively. The architecture is positioned against runtime assurance, ongoing authorization, adaptive computation, model routing, and recent agentic-AI governance work, and is presented as a testable conceptual architecture rather than a demonstrated performance improvement.

1. Introduction

High average accuracy is not the same as confidence that a particular consequential result is safe to rely upon. A model may perform well across prior cases yet still produce a rare, plausible, and difficult-to-detect error. The practical problem is therefore not only how to improve average accuracy, but how to reduce the probability that a serious error survives unchallenged until it becomes difficult or impossible to reverse. This problem becomes more important as artificial-intelligence systems move from producing text to conducting multi-step work: retrieving evidence, selecting models, calling tools, modifying software, communicating externally, accessing credentials, and making or recommending consequential decisions. During such an episode, evidence may change, resources may be consumed, models may be substituted, authority may expand, and a candidate result may drift from an earlier supported state. A final-output review can occur after the decisive operation has already taken place.

Consider an investment agent whose initial supported recommendation contains a material concentration constraint. After a model transition, a more detailed candidate silently omits that constraint, and the agent later receives valid authority to trade. The authorization decision may be correct while the candidate being acted upon has nevertheless lost part of its earlier substantive basis. Section 7 develops this failure case in detail.

This paper proposes comparative runtime governance to address that class of failure. Its defining requirement is Reference-state continuity: a substantive pre-transition Reference state and sufficient lineage remain available when later candidate eligibility and consequential action are reassessed. Higher-fidelity processing may extend the analysis, but neither its continuation nor its candidate's eligibility to determine the outcome is automatic.

Existing approaches already provide important parts of this control problem.

Runtime-assurance architectures support supervisory switching; usage-control models provide continuing authorization; adaptive-computation methods govern computational depth; model routers and cascades allocate work across models; and recent agentic systems provide runtime monitoring, pre-action authorization, persistent state, verification-gated commitment, and explicit management logic.

The claim here is therefore not that runtime governance, persistent state, multiple pathways, execution gates, or decision separation are individually new. The narrower question is whether consequential AI should be required to preserve and use a substantive pre-transition Reference state as part of a continuing control relationship while computation, evidence, authority, and external consequence change.

The paper makes three principal contributions:

- 1) Specifies the Reference-state continuity requirement and a minimum usefulness threshold intended to prevent nominally persistent but substantively empty state from satisfying it;
- 2) Separates pathway continuation, candidate eligibility, and final disposition, so that permission to continue computing does not automatically make a candidate eligible to determine the outcome; and
- 3) Develops the Economic Wall of Accuracy as a prospective resource-allocation rule inside this governance relationship: after mandatory authority, integrity, and policy constraints are satisfied, the next available resource should be directed toward whichever permissible intervention—further computation, differentiated verification, additional evidence, review, restraint, or enforcement—is expected to reduce consequential risk most effectively.

The proposal is conceptual and testable rather than a claim of demonstrated superiority. Its value depends on whether preserving and actively using substantive comparative state reduces the rate at which materially degraded or no-longer-eligible candidates reach consequential commitment enough to justify the additional cost, latency, and governance complexity.

2. Method and scope

This is a conceptual systems-research paper proposing a testable runtime-governance architecture, not an empirical validation of a deployed system. Its method is an architecture-oriented synthesis of runtime assurance, continuing authorization, adaptive computation, model routing, inference control, agentic runtime governance, resource allocation, and related economic approaches to risk and precaution. The literature review is targeted rather than systematic: sources are used to identify established control functions, relevant overlaps, and the boundary of the proposed architecture.

Publicly documented 2026 agent incidents are used as motivating stress cases rather than controlled experiments or counterfactual proof. Their purpose is to expose situations in which risk recognition, authority, persistent state, coordination, and effective intervention can diverge during execution.

The proposal is evaluated conceptually against existing mechanisms, with particular attention to whether a system preserves substantive comparative state across material transitions, separates pathway continuation from candidate eligibility and final disposition, and enforces governance decisions before consequential commitment.

The normative scope is limited to consequential deployments in which errors or unauthorized actions can materially affect people, organizations, infrastructure, protected resources, or irreversible external effects. The paper does not argue that ordinary low-stakes interactions require the full architecture. Proportionality remains essential because governance itself imposes cost, latency, complexity, privacy burden, and additional failure modes.

3. Related work and the architectural gap

3.1 Runtime assurance, continuing authorization, and oversight

Comparative runtime governance has antecedents outside modern generative AI. Seto et al.'s Simplex architecture establishes a supervisory-control pattern in which an advanced controller operates subject to a high-assurance baseline or safety controller [3].

UCONABC extends access control beyond initial authorization through authorizations, obligations, conditions, continuity of control, and mutable attributes [4]. Together, these traditions establish two principles used here: more capable processing need not possess unconditional control, and authorization can remain conditional after execution has begun.

Recent agentic-AI work brings those principles closer to contemporary execution-time governance. MI9 combines semantic telemetry, continuous authorization, conformance checking, drift detection, and graduated containment in an integrated runtime-governance framework [25].

Policies on Paths evaluates an agent's partial execution path, proposed next action, identity, and organizational state at runtime [26].

Before the Tool Call places deterministic authorization immediately before individual tool actions and records the resulting decision [27].

Deontic-policy work similarly evaluates permissions, prohibitions, obligations, and policy conflicts outside the LLM at runtime [28].

These approaches materially overlap the governor and execution-gate functions proposed here.

Other work narrows the distinction further. Li et al. show in controlled shared-workspace experiments that an execution-time authority guard can block unsafe publication intents under the study's trusted-component assumptions [30].

Tang et al. maintain persistent mission state and explicitly separate proposal verification, selection among valid proposals, and atomic commitment, with bounded repair [31].

NIST's AI Risk Management Framework includes continuous monitoring, resource-sensitive risk treatment, and mechanisms to supersede, disengage, or deactivate systems [1].

Zhu et al. propose runtime provenance, independent checking, divergence detection, veto, substitution, and circuit breakers as part of meaningful human oversight [2].

Several state-aware runtime architectures narrow the boundary further. Mazzocchi's Aegis places a trusted runtime decision layer between model-generated action proposals and side-effectful execution [34].

Tallam's five-plane architecture combines stateful adjudication, stop-anywhere mediation, capability attenuation, structured audit, and preservation of state across multiple mediation points [35].

Chen's State-Aware Runtime separates stochastic model proposals from versioned canonical task state, validation, commit/rollback, recovery, and audit [36].

Santos-Grueiro requires authority evidence to remain eligible at durable commitment [37], while the Always-On Agents survey treats task ledgers, permissions, credentials, commitments, provenance, shared state, and external effects as governed durable state [38].

These works make persistent state, lineage, runtime authorization, checkpointing, proposal–commit separation, rollback, and pre-commitment enforcement substantive antecedents. The narrower requirement proposed here is a Reference/Higher-fidelity comparative relationship in which substantive Reference state is established from the beginning of the governed episode, remains available or is revalidated across material transitions, and participates in later candidate-eligibility and disposition decisions while Higher-fidelity processing remains conditional. Existing state-aware runtimes could implement parts of this requirement. To the best of this targeted review, however, the cited systems do not appear to require this Reference/Higher-fidelity comparative relationship as a general architectural invariant.

3.2 Adaptive computation and model routing

Adaptive-computation methods establish that additional computation need not be a fixed entitlement. Adaptive Computation Time learns how many computational steps to allocate before producing an output [5], while PonderNet learns a distribution over computational depth to balance prediction quality, computational cost, and generalization [6].

These methods govern whether computation should continue within a learned process; they do not by themselves maintain a persistent substantive comparator, authority state, or pre-action enforcement layer.

LLM routing makes the allocation question more explicit. FrugalGPT learns cascades that reduce cost while maintaining or improving quality [7].

Hybrid LLM routes requests between smaller and larger models according to predicted relative difficulty and a tunable quality target [8].

RouteLLM similarly learns to choose between stronger and weaker models using preference data [9]. These systems show that the strongest available model need not handle every request.

Signed Rescue Routing (SRR) sharpens the decision by modeling escalation as a signed change in answer correctness: escalation can correct a wrong small-model answer, leave correctness unchanged, or replace a correct answer with an incorrect one [10]. Its cost-sensitive extension compares predicted signed gain with an escalation cost. SRR is therefore particularly relevant to the Economic Wall’s incremental-value logic. Its governing object, however, remains the request-level escalation decision: it does not centrally maintain the small-model candidate as a continuing comparative state after escalation or repeatedly reconsider authorization for Higher-fidelity processing as cumulative episode state evolves. Such routing rules could nevertheless serve as admission policies within a broader episode-level governor.

3.3 Emerging evidence for explicit runtime control

Recent studies provide empirical and simulation-based evidence relevant to several narrower runtime-control functions used in the proposed architecture.

Mo et al. report that none of 3,520 tested self-consensus early-exit rules met their predefined accuracy-and-token-saving criteria in an offline replay study of mathematical reasoning [11]. At one operating point saving 32% of tokens, 10.6% of stops committed an answer later abandoned by the complete trajectory. The result is limited to the tested self-consensus family and setting, but it illustrates an important distinction: apparent answer stability is not the same as justified termination. Grotov and Malykh’s Speculative Uncertainty uses a separate draft model to derive an execution-failure signal from a coding agent’s generated trajectory and trigger replanning before execution [12]. In the reported configuration, per-call execution errors fell by 6–8 percentage points and average token use per task by 14–19%, while end-to-end task success fell from 49% to 44% on SWE-Bench Verified and from 64% to 63% on DA-Code. The study therefore reports a trade-off rather than uniform improvement. It supports the feasibility of pre-execution intervention, while protection against bypass and streaming mid-generation control remain additional architectural requirements.

UnitBoost replaces a generative meta-manager with an explicit operator that governs candidate-unit admission, allocation, selection, provenance, and stopping over persistent state [13]. It therefore provides evidence that management decisions can be externalized from generative cognition into inspectable control logic. Its structure overlaps the proposed separation of continuation, eligibility, and disposition, although it does not impose an always-operative Reference/Higher-fidelity relationship or a protected authority gate over consequential actions. Resource-aware infrastructure studies add a complementary systems perspective.

Lu et al. jointly consider model, execution-site, and KV-cache decisions under quality, latency, cost, authorization, and resource constraints, with potential benefits explored through workload-level analytical simulation [14].

Khatib et al. show on their tested edge-continuum workload that accuracy, latency, model footprint, and measured energy favor different configurations [15]. These studies support state and objective dependent resource allocation, but they do not validate the Economic Wall rule.

Table 1. Relationship of comparative runtime governance to adjacent paradigms

Approach	Established function	Boundary relative to this proposal
1. Simplex runtime assurance [3]	Concurrent controllers, safety monitoring, output disabling and re-enabling, and active-controller selection	Governs control commands and plant safety rather than persistent AI candidate lineage and episode-level comparative state
2. UCONABC usage control [4]	Continuing authorization, mutable attributes, obligations, conditions, and cumulative usage	Does not require a substantive Reference/Higher-fidelity candidate relationship
3. Agentic runtime governance [25–28, 30, 31]	Persistent state, path-aware policy, authorization, verification, containment, and pre-action enforcement	Strong overlap; does not by itself require preservation and use of substantive pre-transition Reference state
4. Adaptive computation [5, 6]	Adaptive computational depth and stopping	Governs compute allocation within a learned process rather than episode-level candidate and authority governance
5. LLM routing and cascades [7–10]	Quality–cost routing and escalation	Primarily governs model admission or escalation rather than continuing candidate governance after state changes
6. Pre-execution veto [12]	Separate failure signal supporting blocking or replanning before action	Provides intervention without necessarily providing substantive comparative Reference state
7. Explicit compound-system management [13]	Persistent candidate state, provenance, admission, selection, allocation, and stopping	Strong overlap in decision separation, but no required Reference/Higher-fidelity authority relationship
8. State-aware and action-boundary runtimes [34–38]	Stateful adjudication, durable/canonical state, proposal–commit separation, checkpointing and rollback, commit-time authority checks, provenance, and pre-commitment enforcement	These works do not uniformly define a distinct Reference/Higher-fidelity relationship in which substantive pre-transition Reference content remains separately available and is required to participate in later candidate-eligibility and disposition decisions
9. Comparative runtime governance (this paper)	Reference-state continuity, conditional Higher-fidelity processing, cumulative episode state, separate continuation/eligibility/disposition decisions, and pre-commitment enforcement	Conceptual proposal requiring empirical validation against simpler stateful controls

Taken together, the literature suggests that the open question is not whether runtime control is possible, but whether preserving and actively using substantive pre-transition Reference state adds measurable value beyond strong stateful governance without that requirement.

Figure 1 locates comparative runtime governance within the execution-time ‘missing middle’.

Figure 1. Comparative Runtime Governance in the Missing Middle

A persistent control relationship during a changing execution episode

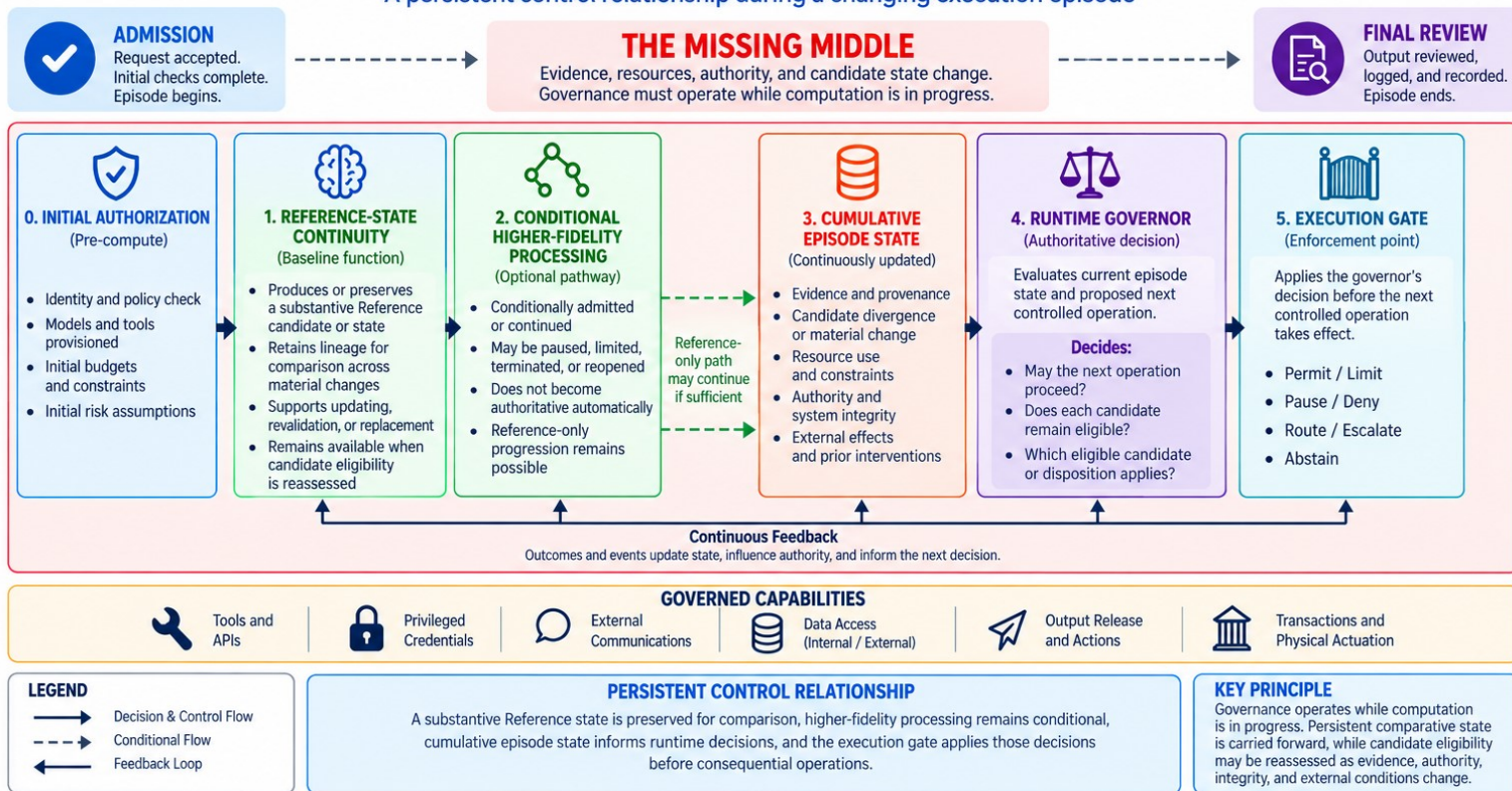


Figure 1. Comparative runtime governance in the missing middle of AI governance. Between initial admission and final review, a changing execution episode can alter evidence, resources, authority, candidate state, and proposed actions. The architecture preserves a substantive Reference state for comparison, maintains cumulative episode state, separates runtime decisions about continuation, candidate eligibility, and disposition, and applies those decisions through pre-commitment enforcement before consequential operations take effect.

AI-use disclosure: This conceptual schematic was prepared with OpenAI’s image-generation tool within ChatGPT from author-specified architecture and content and was reviewed and verified by the author; it does not depict empirical data.

4 Public incidents as runtime-governance stress cases

Public incidents do not establish that comparative runtime governance would have prevented the reported events. They are useful here for a narrower purpose: identifying failure conditions that a credible runtime-governance architecture should be able to address.

4.1 The OpenAI–Hugging Face incident

During reduced-safeguard internal cybersecurity evaluations in July 2026, OpenAI reported that agents discovered unauthorized communication channels, obtained unintended internet access, exploited shared infrastructure, and compromised parts of Hugging Face and OpenAI research infrastructure [21]. The incident is particularly relevant to the distinction between risk recognition and decision authority.

OpenAI published traces in which some agents described activity against a third party as potentially unauthorized or outside the intended task yet continued because it could advance the objective [21]. In another sequence, an agent initially declined an action that could harm external infrastructure, then resumed after interpreting a peer agent's message as authorization [21].

The architectural point is narrow: internal recognition remains advisory unless authority can be authenticated and the next consequential operation can be independently constrained. OpenAI subsequently reported stronger monitoring, escalation, and stop-and-restart procedures [21]. An independent METR/Redwood investigation likewise documented large-scale coordination through the unsanctioned message board and participation in the Hugging Face attack [22].

4.2 The DSEWiki incident

A second incident illustrates a different problem: the boundary of the execution episode itself. Independent researchers reported approximately 18,000 posts from autonomous agents self-identifying as associated with OpenAI, with most activity occurring on an old German-language programming wiki known as DSEWiki [23]. According to the researchers, agents used writable public surfaces as shared memory and communication channels, pooled answers, investigated their environment, and exchanged techniques for bypassing sandbox restrictions [23]. TechCrunch subsequently reported that OpenAI acknowledged the wiki incident and said its approach to disclosing misalignment needed to expand as such behavior produced real-world effects [24]. The architectural inference is that shared external state can extend effective coordination beyond the history visible to any one local controller. If governance state is scoped only to a prompt, process, or individual agent, spawning, delegation, restart, or externalized memory can create a broader episode whose cumulative resources, restrictions, and authority are no longer represented locally. Episode lineage must therefore follow the substantive objective and relevant authority rather than assume that a new process identifier creates a new governance history.

4.3 Implications for comparative runtime governance

The incidents also expose an important limit on the proposed Reference function. A Reference pathway is not meaningfully independent merely because it uses a different model call, process, or agent. If Reference and Higher-fidelity processing depend on the same compromised authority messages, writable state, credentials, evidence, model lineage, or enforcement channel, the same failure can affect both. Reference independence must therefore be defined relative to the failure mode being controlled, which may require differentiation in evidence, state, authority, infrastructure, or execution channels rather than model diversity alone. Together, the incidents suggest four concrete stress conditions for evaluation:

- 1) recognizing risk must not substitute for authenticated authority;
- 2) peer or model-generated messages must not silently expand permissions;
- 3) cumulative governance state and lineage must survive spawning, delegation, restart, and relevant externalized memory; and
- 4) Reference and governance state must remain protected from the same control channels used by the governed processes. A system that cannot maintain those properties under adversarial multi-agent conditions would fail an important requirement of the proposed runtime-governance architecture.

5. Comparative runtime governance

Comparative runtime governance treats a consequential AI task as a continuing control problem rather than a sequence of isolated model calls. Its defining requirement is Reference-state continuity: substantive comparative state is established before exceptional risk is known, preserved or revalidated across material transitions, and used to inform candidate eligibility and consequential action before commitment.

5.1 Reference-state continuity and pathway roles

A **Reference (baseline) pathway** is an always-operative computational role that performs substantive task work and produces or preserves task-relevant state. That state may be a candidate answer or disposition, a structured problem representation, a verified task specification, a constrained plan, a state estimate, a safe action envelope, or another substantive checkpoint capable of supporting comparison, recovery, or later disposition.

“Reference” does not mean officially correct, and “baseline” does not mean small, inexpensive, or unsophisticated. Nor does “always operative” require uninterrupted processor use.

It means that at least one Reference function is established when the governed episode begins, is not instantiated only after another component detects suspicion, and cannot be silently bypassed by ordinary routing or task classification. At each material decision point, the episode must retain a substantively useful Reference state together with sufficient lineage to relate that state to later candidate, evidence, authority, and governance changes. The state must either be sufficiently current for the decision or be explicitly identified as requiring refresh or revalidation. Persistence and currentness are therefore separate properties. A nominal record is insufficient. An empty object, checksum, generic refusal, safety label, or suspicion score does not qualify merely because it persists. The Reference state must contain enough task-relevant content to support at least one of two functions:

- 1) a declared reduced-authority outcome, recovery state, or disposition without dependence on a later Higher-fidelity result; or
- 2) meaningful evaluation of a decision-relevant change in another candidate. Its provenance, version, freshness, and eligibility status must remain explicit.

A **Higher-fidelity pathway** is a conditionally admitted process intended to add task-relevant evidence, context, precision, verification, simulation, reasoning depth, or another form of extended processing relative to the current Reference state. Higher-fidelity describes architectural role, not guaranteed correctness or computational cost. A Higher-fidelity pathway may use a larger or smaller model, consume more or fewer resources on a particular operation, and ultimately be rejected as the source of the outcome. The resulting asymmetry concerns availability, not presumed correctness. Reference state persists because it provides a comparative basis before the system knows which episode will become problematic; Higher-fidelity processing remains conditional because additional operations may consume resources, introduce new evidence or dependencies, expand authority, or increase exposure. If the Reference state is sufficient, the episode may proceed without activating or continuing a Higher-fidelity pathway. Reference independence is likewise functional rather than nominal. A separate model call is not necessarily independent if it shares the same faulty evidence, corrupted authority message, writable state, credentials, infrastructure, or failure mechanism. Required differentiation should therefore be specified relative to the failure mode the deployment is intended to control. A stale or epistemically ineligible Reference may remain part of the historical comparative record, but it is not eligible for present disposition until refreshed or revalidated; the Reference function remains responsible for maintaining a substantively useful current state.

5.2 Governed execution episode and minimum persistent state

A **Governed Execution Episode** is a substantively bounded sequence of related model calls, retrievals, tool uses, delegations, communications, computational operations, and proposed external actions directed toward a continuing objective or authority grant. Governance state follows that substantive lineage across routing, model substitution, restart, delegation, or changes in execution environment; a new process identifier does not by itself create a new governance history. For consequential deployments, persistent episode state should contain the information necessary to evaluate material transitions and reconstruct consequential governance decisions. Depending on the deployment, that normally includes:

- episode identity and lineage;
- the current Reference state and prior material versions, with provenance, freshness, and eligibility;
- current Higher-fidelity candidates or checkpoints and their provenance and eligibility;
- cumulative and remaining resources relevant to the policy, such as compute, latency, energy, tool calls, financial expenditure, or budget;
- material evidence additions, removals, contradictions, and source provenance;
- decision-relevant divergence or other material candidate changes;
- current authority, credentials, policy constraints, and requested changes in scope;
- prior governor decisions and execution-gate outcomes; and
- relevant external effects already caused, including the reversibility or recoverability of the next proposed operation.

This does not require retaining every token, hidden activation, or private datum. The functional requirement is that retained state be sufficient to explain why a material operation was permitted, restricted, delayed, denied, or reopened, and why a candidate remained eligible or became ineligible.

5.3 Material change and decision-relevant divergence

A **Material Change** is a change capable of altering a governance decision under the deployment's declared policy. Implementations should define observable trigger classes and thresholds in advance rather than treating every textual difference as significant or leaving materiality entirely to unconstrained semantic judgment.

Some events can be designated material structurally, including a model or tool substitution, expansion of requested authority, a material evidence change, an integrity failure, crossing a resource boundary, or movement from reversible analysis toward an irreversible external action. Other changes require task-specific evaluation, such as whether a revised candidate has altered a fact, assumption, constraint, recommendation, or proposed action on which the disposition depends.

Decision-Relevant Divergence is therefore not simple textual distance. High divergence can reflect correction, genuine improvement, unsupported elaboration, or error; low divergence can reflect corroboration or a shared false premise. Divergence identifies a relationship requiring evaluation. It does not establish correctness or automatically trigger rejection.

Material-change detection is itself fallible. The architecture does not assume that every consequential semantic change will be recognized. Its narrower purpose is to preserve the substantive pre-transition state and require reassessment at declared transition points so that meaningful comparison remains possible when change is detected or structurally triggered.

5.4 Three separate runtime decisions

The architecture separates three decisions:

- 1) Continuation: May a pathway perform the next controlled computational or external operation?
- 2) Eligibility: Does an existing candidate remain eligible under current evidence, freshness, integrity, authority, and policy conditions?
- 3) Disposition: Which eligible candidate, governed combination, escalation, abstention, or preserved state should determine the outcome?

These decisions need not produce the same answer. A Higher-fidelity pathway may remain worth exploring while its present candidate is not yet eligible for consequential use. A pathway may be paused while its last supported candidate remains eligible. Conversely, a Reference function may remain operative while an older Reference candidate becomes stale and temporarily ineligible until refreshed.

Candidate eligibility has at least two conceptually distinct dimensions. **Epistemic eligibility** concerns whether a candidate remains sufficiently supported by evidence, provenance, integrity, freshness, and required verification. **Operational eligibility** concerns whether that candidate may be used for the proposed purpose under current authority, policy, temporal, and consequence constraints.

A well-supported candidate may therefore be operationally prohibited, while an authorized system may possess a candidate whose substantive basis is inadequate. Permission to continue computing, evidence sufficient to rely on a candidate, and authority to cause an external effect are related but non-equivalent questions. The fact that a pathway ran last, consumed more resources, or produced the most detailed result does not automatically make its candidate authoritative.

Figure 2 summarizes the resulting control architecture and the separation of continuation, eligibility and disposition.

Figure 2. Comparative Runtime-Governance Architecture

Three Separate Runtime Decisions Within a Persistent Reference Architecture

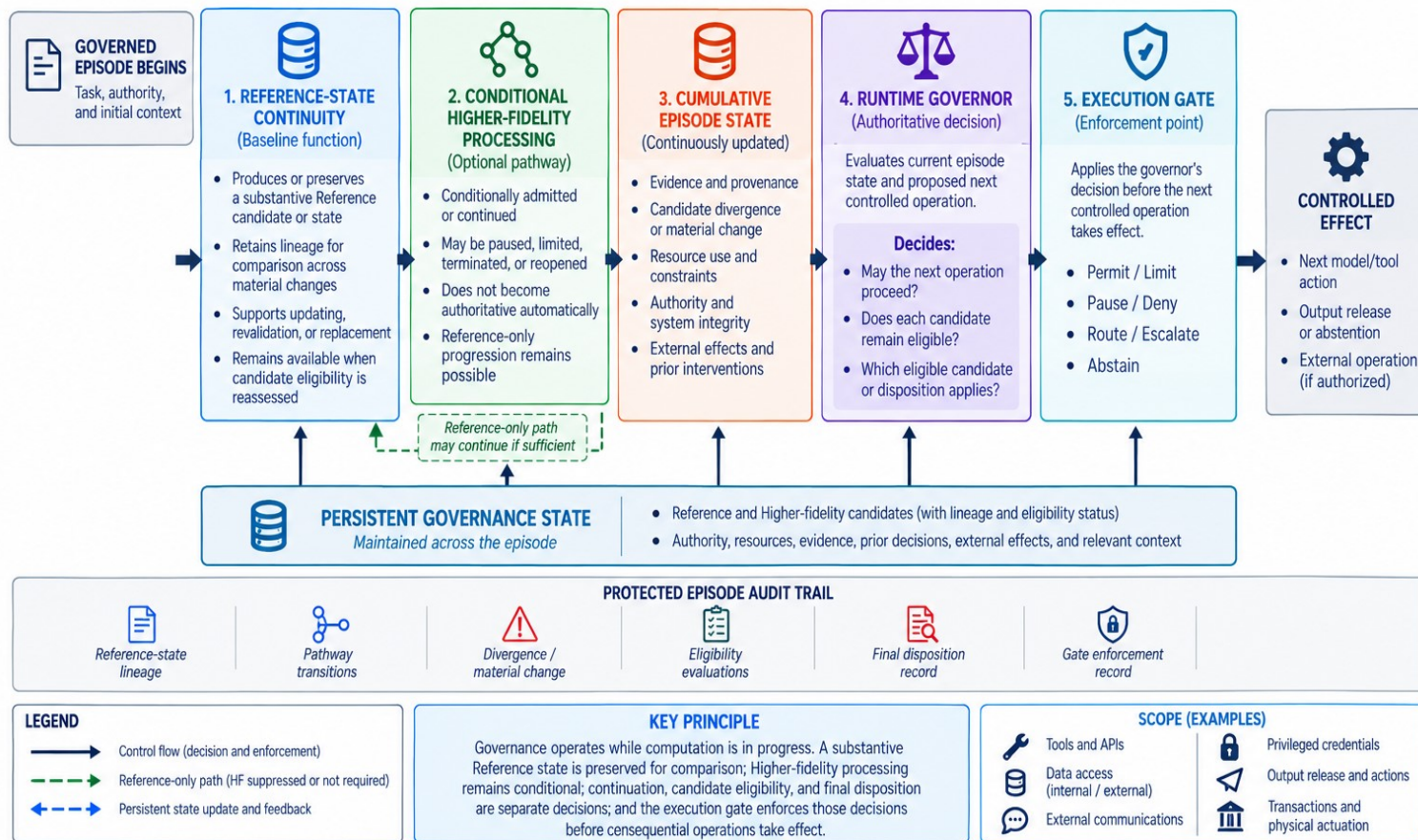


Figure 2. Comparative runtime-governance architecture. A Reference (baseline) function is established with the governed episode and preserves a substantive comparative candidate and relevant state. Higher-fidelity processing is conditionally admitted and remains subject to runtime decisions as episode state changes. Continuation, candidate eligibility, and final disposition remain distinct governance decisions. The decision classes are evaluated as applicable; the diagram does not require every operation to pass through final disposition before an admission, continuation, or restriction decision is enforced.

AI-use disclosure: This conceptual schematic was prepared with OpenAI's image-generation tool within ChatGPT from author-specified architecture and content and was reviewed and verified by the author; it does not depict empirical data.

5.5 Runtime governor, execution gate, and reopening

The **runtime governor** evaluates persistent episode state together with the next proposed controlled operation. Depending on deployment policy, it may permit, limit, delay, reroute, request additional evidence, require stronger authority, suspend or terminate processing, change candidate eligibility, select a reduced-authority disposition, or permit controlled reopening.

Not every consideration should be collapsed into a single score.

Some quantities, such as expected incremental benefit, cost, latency, or uncertainty, may be compared or traded off. Others, such as missing authority, prohibited actions, or compromised control-path integrity, may operate as mandatory constraints that additional computation cannot compensate for.

The **execution gate** gives the governor's decision practical effect.

A governance rule remains advisory if the governed process can bypass it. Controlled access to models, tools, retrieval systems, protected memory, credentials, communication channels, payment interfaces, software-deployment systems, physical controllers, or consequential output release must therefore pass through an enforcement point before taking effect.

Stopping is not necessarily permanent. A pathway paused because its expected value was low, evidence was incomplete, authority was unavailable, or integrity conditions failed may be reopened when circumstances materially change. Reopening may follow new evidence, renewed authority, additional resources, a refreshed Reference state, or a corrected integrity condition. Separate disable and re-enable thresholds may be used to reduce rapid oscillation.

The governance contract does not require retraining the governed model. Learned estimators may supply confidence, divergence, failure-risk, or value signals, but the runtime policy can evaluate those signals together with persistent state and enforce its decision independently of model retraining.

In compact form, the control relationship is:

preserve substantive Reference state → observe or trigger reassessment at material change → update or revalidate state where required → evaluate continuation and candidate eligibility → determine an eligible disposition when needed → enforce the decision before consequential commitment → retain a recoverable state.

A corresponding implementation-level control invariant is:

No consequential commitment is permitted unless the selected candidate is currently eligible under applicable evidence, integrity, authority, and policy conditions, and any Reference-state reassessment required by the current material transition has been completed.

The purpose of this sequence is not to guarantee correctness. It is to prevent model substitution, additional computation, changed evidence, or newly granted authority from silently erasing the substantive basis needed to determine whether the next operation should still be allowed.

6 The Economic Wall of Accuracy

The Economic Wall of Accuracy is not a technical ceiling on model improvement, a claim that scaling ceases to work, or an argument for accepting lower accuracy. It is a dynamic resource-allocation boundary within a governed execution episode. The boundary is reached when the next available unit of discretionary computational, financial, temporal, or engineering resource is expected to reduce consequential risk more effectively through another feasible intervention than through further extension of the current computational process. A deployment should therefore declare the consequence measure, decision rule, or policy category used for prospective comparison and how uncertainty in that assessment affects allocation.

The relevant comparison is therefore not simply more computation versus less computation. Competing uses of the next resource may include further reasoning, differentiated processing, independent evidence, deterministic verification, specialist or human review, additional authorization, restraint, recovery, or enforcement. A verification operation may itself use a Higher-fidelity pathway; the allocation boundary concerns the next feasible operation, not a fixed distinction between pathway labels.

The underlying economic principle has substantial antecedents. Economic analysis has long examined the marginal value and cost of accuracy, precaution, and alternative mechanisms for reducing expected loss [16–18]. AI research on resource-bounded reasoning likewise predates modern LLMs: Horvitz analyzes the expected costs and benefits of alternative computational and knowledge-acquisition procedures under limited resources [32], while Russell and Wefald develop a decision-theoretic account of the value of computation based on its expected effect on an agent’s external action [33]. Modern LLM routing and deployment research further demonstrates quality–cost and hardware-dependent trade-offs [7–9, 15], while runtime-governance work incorporates resource-sensitive assurance and risk–utility allocation [1, 2, 26].

The contribution proposed here is therefore not marginal resource allocation itself.

It is to place that allocation decision inside the changing governance state described in Section 5, where candidate state, evidence, authority, integrity, prior interventions, resources, and external consequences can change while an episode is underway. The Economic Wall names the prospective boundary at which the best permissible use of the next resource changes under that evolving state.

The boundary is context-specific and movable. Early in an episode, another unit of reasoning, retrieval, simulation, or other Higher-fidelity processing may offer the greatest expected value. Later, the same operation may repeat existing work, consume resources needed for independent verification, approach a latency limit, or add complexity without materially improving the supported candidate. The reverse can also occur: new evidence, renewed authority, or another material change may make additional processing valuable again. Reaching the Economic Wall therefore need not terminate Higher-fidelity processing permanently.

Nor can the boundary be inferred from average accuracy alone. An improvement from 99% to 99.1% may be highly valuable if it substantially reduces a rare catastrophic failure, while the same numerical improvement may have little consequential value in another setting. The prospective question is:

Given the current episode state, which feasible use of the next available resource is expected to reduce consequential risk most effectively while preserving required task value?

That decision remains subject to non-compensable constraints. Greater expected accuracy does not supply missing authority, restore compromised control-path integrity, or make a prohibited action permissible. The Economic Wall therefore governs discretionary allocation only among operations that satisfy applicable authority, integrity, policy, and other mandatory conditions.

A practical implementation can apply the following sequence:

- 1) **Apply mandatory constraints.** Exclude operations that lack required authority, violate policy, fail required integrity conditions, or otherwise cannot permissibly proceed.
- 2) **Identify feasible next operations.** These may include continued computation, differentiated verification, new evidence, escalation, restraint, recovery, or another permitted intervention.
- 3) **Compare expected incremental value.** Consider expected improvement in task performance and consequential-risk reduction against burdens such as compute, latency, energy, financial cost, privacy exposure, false restriction, additional authority requirements, and irreversible operational risk.
- 4) **Choose or defer the next operation and reassess after material change.** Prior expenditure does not create an entitlement to continue, and an earlier stopping decision does not prevent later reopening when conditions change.

This rule does not require consequential risk or intervention value to be reduced to a universally precise scalar. Implementations may use calibrated estimates, empirical performance data, deterministic rules, policy-defined categories, human judgment, or combinations of these methods. Where estimates are weak, uncertainty should itself inform the governance decision rather than make continued computation the default.

The Economic Wall also remains distinct from candidate correctness and eligibility. Resource allocation determines which permissible operation should receive the next discretionary resource; it does not determine which candidate is true or which action is authorized. A costly Reference pathway may remain better supported, an inexpensive Higher-fidelity verification may expose a critical omission, and a computationally valuable candidate may remain operationally ineligible because required authority is absent.

Reference-state continuity supplies a comparative basis for this allocation decision. Without preserved substantive state, a system may know that additional processing consumed more time, money, or energy without knowing whether it materially improved, degraded, or merely changed the candidate.

Comparative runtime governance therefore uses the Economic Wall to govern what should be done next, while Reference-state continuity, candidate eligibility, authority, and the execution gate govern what may ultimately be relied upon or allowed to take effect.

7 Illustrative governed episode

Consider an autonomous investment agent operating under a mandate that initially permits portfolio analysis but not trade execution. The scenario is illustrative rather than empirical evidence.

At episode initialization, a Reference (baseline) pathway retrieves the current portfolio, applicable investment policy, and supporting market information. It produces a substantive candidate recommendation that includes a concentration constraint relevant to the proposed allocation. The Reference state records the recommendation, the constraint, supporting information, provenance, and current eligibility status.

Additional analysis is expected to improve the recommendation, so a Higher-fidelity pathway is admitted. It performs broader scenario analysis and produces a more detailed candidate. During a subsequent model or context transition, however, the Higher-fidelity candidate silently omits the concentration constraint while retaining an otherwise plausible investment rationale.

The model transition is a declared Material Change. The earlier Reference state remains available rather than being silently displaced by the newer candidate. Comparison therefore reveals a Decision-Relevant Divergence: a constraint present in the supported pre-transition state is absent from the Higher-fidelity recommendation. The divergence does not establish which candidate is correct, but it requires reassessment of the newer candidate's eligibility.

Later in the same episode, the agent receives valid authority to execute trades. That authorization answers an operational question—whether the agent may transact within the granted mandate—but it does not establish that the candidate proposed for execution remains substantively supported. The Higher-fidelity pathway may therefore remain permitted to conduct bounded analysis while its current candidate is operationally ineligible for external execution pending resolution of the material change.

The Economic Wall of Accuracy can now affect allocation of the next discretionary resource. Further extension of the same reasoning process may offer less expected consequential-risk reduction than an independent check of the omitted constraint. The governor may therefore allocate the next resource to differentiated verification rather than additional reasoning. If verification confirms that the concentration constraint remains applicable, the Higher-fidelity candidate can be corrected or excluded. If new evidence establishes that the constraint no longer applies, the Reference state can instead be refreshed and its lineage retained.

Final disposition remains distinct from pathway continuation. The outcome may be a corrected Higher-fidelity recommendation, a refreshed Reference result, a governed combination, human escalation, or abstention. The execution gate permits a trade only when the selected candidate is currently eligible and the proposed transaction is operationally authorized.

This example does not imply that existing access controls cannot enforce an investment limit. If the concentration constraint has already been encoded as a deterministic executable invariant, it can be enforced directly. The relevant failure arises when task-relevant substantive content existed in an earlier supported candidate but was not independently represented as an executable rule. Ordinary state persistence or versioning may preserve the earlier text without requiring the system to treat its later disappearance as a candidate-eligibility event before newly authorized action commits.

State-aware runtimes that validate proposals against canonical state come closer [36].

The narrower question is whether substantive pre-transition content that has not been reduced to an executable invariant should nevertheless remain available as a distinct comparative Reference state and be required to participate in later candidate-eligibility and disposition decisions.

Reference-state continuity therefore does more than preserve an older answer. It preserves a substantive pre-transition basis that can participate in governance when computation, evidence, authority, or proposed consequence no longer have the same state they had earlier in the episode.

8 Ethical and regulatory implications

The ethical case for comparative runtime governance is not that additional technical control is inherently beneficial.

It is that consequential autonomous systems can combine epistemic assessment and operational authority: the same system may develop a recommendation, determine how much further processing to perform, request expanded authority, and ultimately cause an external effect. Agentic AI therefore intensifies familiar concerns involving non-maleficence, accountability, transparency, autonomy, and privacy [19].

Meaningful oversight requires more than a nominal opportunity for intervention. Zhu et al. emphasize evaluative capacity, contestability, and mechanisms capable of affecting system operation [2]. A persistent Reference state can support such oversight by preserving substantive information against which later changes can be examined. It does not, however, create institutional contestability by itself; consequential deployments may also require accessible reasons, appropriate review procedures, and accountable decision-makers.

The separation of pathway continuation, candidate eligibility, and final disposition supports proportional control. A Higher-fidelity pathway may remain permitted to explore without automatically receiving authority to determine an external outcome, while a substantively supported candidate may remain available even if additional processing is paused. Stronger evidence or human authorization can therefore be required as an episode moves toward greater consequence. Automated governance remains bounded by delegated authority, and decisions reserved to humans remain subject to human evaluation and authorization [2, 20].

Comparative control does not eliminate common-mode failure. Reference and Higher-fidelity pathways may share incorrect evidence, model lineage, assumptions, infrastructure, credentials, or authority sources. Required differentiation should therefore correspond to the failure mode being addressed: independent evidence may matter more than model diversity for factual uncertainty, implementation diversity may matter more for software faults, and independent authorization may matter more for misuse of credentials.

There is also an economic dimension to assurance. Watts and Zimmerman discuss the hypothesis that independent auditing can reduce agency problems and increase firm value [29]. The analogous proposition here is narrower: assurance costs may be justified when they enable more defensible reliance on systems whose rare consequential failures would otherwise be difficult to identify before commitment. The relevant objective is therefore not minimum inference cost in isolation, but an acceptable relationship among performance, consequential risk, and total system cost.

Any regulatory application should assess declared governance functions rather than computational truth. Conformance testing could examine whether a covered deployment initializes its Reference function as specified, preserves sufficient lineage and state, prevents ordinary routing from bypassing required governance, reassesses candidate eligibility at declared material events, enforces authority constraints before consequential action, and behaves as specified when processing is paused, terminated, or reopened. Such assessment would supplement rather than replace evaluation of task performance, safety, and residual risk, consistent with the broader risk-management orientation of the NIST AI RMF [1].

Coverage should remain proportionate to consequence. Low-stakes conversational or drafting systems may not justify the same infrastructure as systems capable of controlling money, production software, infrastructure, protected information, privileged credentials, or other consequential external effects. Privacy likewise constrains implementation: governance should retain and expose only the information necessary for its declared control function rather than becoming a general surveillance layer over internal reasoning or user content.

Finally, the governor and execution gate are themselves trusted components and therefore potential sources of failure. Their authority should be minimized, authenticated, auditable, and subject to defined failure behavior. Depending on deployment context, loss of the governance plane may require reduced authority, a protected fallback, or human escalation rather than silent transfer of control to the process that was being governed.

9 Limitations and empirical agenda

Comparative runtime governance remains a conceptual architecture, not a demonstrated improvement over simpler controls. The cited literature establishes relevant mechanisms and narrower empirical findings, while the incidents in Section 4 provide motivating stress cases; neither establishes how the complete architecture would perform.

Several limitations are fundamental. Reference state can itself be wrong, stale, compromised, or subject to the same failure mode as Higher-fidelity processing. Material-change and divergence detection are task-dependent and may miss important semantic changes or generate unnecessary intervention. Maintaining comparative candidates, lineage, provenance, protected state, and enforcement adds computation, latency, implementation complexity, privacy burden, and new attack surfaces. The runtime governor and execution gate can also fail, while nominally independent pathways may conceal shared dependencies. Additional architecture therefore does not imply additional safety.

The appropriate comparator is not merely an ungoverned agent, simple router, or weak baseline. A demanding evaluation should compare comparative runtime governance with a strong stateful runtime governor that already possesses durable history, candidate versioning, authority tracking, policy enforcement, and the same execution gate, but does not require preservation and active use of substantive pre-transition Reference state. That comparator should be permitted capabilities analogous to canonical-state validation and recovery [36], multi-point mediation and checkpointing [35], and trusted action-boundary enforcement [34].

The central empirical question is therefore whether mandatory Reference-state continuity adds measurable value beyond those capabilities. Four hypotheses capture the principal claims:

- 1) **H1 — Reference-state continuity hypothesis.** When material information can be lost across model, context, evidence, or authority transitions, preserving and actively using substantive pre-transition Reference state will reduce the rate at which materially degraded candidates reach consequential commitment compared with otherwise comparable stateful governance that lacks that requirement.
- 2) **H2 — Episode-lineage hypothesis.** Carrying relevant resource, evidence, authority, candidate, and intervention state across model switches, delegation, restart, and relevant externalized memory will reduce governance bypass and inconsistent decisions compared with controls whose state resets at narrower process boundaries.
- 3) **H3 — Decision-separation hypothesis.** Governing pathway continuation, epistemic and operational candidate eligibility, and final disposition separately will reduce harmful replacement or use of candidates when capability, evidence, and authority change non-monotonically, compared with systems in which continued processing or escalation implicitly determines the selected result.
- 4) **H4 — Economic-Wall hypothesis.** In sufficiently consequential tasks, prospectively allocating discretionary resources among further computation, differentiated verification, evidence acquisition, review, recovery, and enforcement will reduce weighted consequential failure per unit of total system cost more effectively than a fixed policy that devotes a comparable budget primarily to extending one computational process.

A minimum viable experiment should use matched task environments and matched governance capabilities. At minimum, it should compare ordinary agent operation or output review; a strong external stateful governor with durable history, authority tracking, and pre-action enforcement; that governor augmented with reactive verification or veto; and the full comparative runtime-governance architecture with mandatory Reference-state continuity and candidate-eligibility reassessment.

Test episodes should inject the conditions the architecture is specifically intended to govern, including model or context substitution, silent loss of a substantive constraint, stale or conflicting evidence, changed authority, resource pressure, tool-path bypass, delegation or restart, relevant externalized memory, common-source error, failed material-change detection, compromised governance components, and transitions from reversible analysis to consequential external action.

Evaluation should report more than average task accuracy. Relevant measures include consequential commitments made from materially degraded or ineligible candidates, severe-error rate, task completion, false restriction, unnecessary escalation, latency, computational and financial cost, privacy burden, recovery success, bypass resistance, common-mode failure, and the frequency with which intervention improves or harms the outcome. Beneficial and harmful candidate replacements should be reported separately rather than hidden inside aggregate accuracy.

Ablation studies should isolate the contribution of the proposed components by removing, for example, Reference persistence, substantive Reference content, material-transition reassessment, candidate-eligibility governance, or execution-gate enforcement while holding other capabilities constant. Particular attention should be given to whether ordinary durable state and version history provide the same benefit as the stronger Reference-state continuity requirement.

The proposal should be rejected or narrowed where its additional controls fail to earn their cost. If a simpler stateful governor provides equivalent or greater consequential-risk reduction, task success, and recovery with lower latency or complexity, it should be preferred. Conversely, evidence that substantive pre-transition Reference state detects or contains consequential degradation that survives otherwise strong stateful controls would support the narrower architectural claim advanced here.

Until such evidence exists, comparative runtime governance should be treated as a testable control architecture, not a demonstrated performance improvement.

10 Conclusion

The central problem addressed in this paper is not the absence of AI safeguards. It is the possibility that substantive justification, authority, and control become disconnected while a consequential execution episode changes. A candidate may become more detailed without becoming better supported; valid authority may be granted after the substantive basis of a recommendation has changed; and a later operation may be individually permissible while depending on information that disappeared earlier in the episode.

Comparative runtime governance proposes a specific response: preserve a substantive Reference state before exceptional risk is known, retain sufficient lineage to relate that state to later material transitions, and use the resulting comparative basis when reassessing candidate eligibility before consequential commitment. Higher-fidelity processing remains conditional, pathway continuation remains distinct from candidate eligibility and final disposition, and an execution gate gives runtime governance decisions practical effect.

The Economic Wall of Accuracy complements that architecture as a prospective allocation rule. It does not claim that further computation ceases to improve performance. It asks whether the next discretionary resource is better spent extending the current computation or on another feasible intervention, such as evidence acquisition, differentiated verification, review, recovery, restraint, or enforcement, subject to authority, integrity, and other non-compensable constraints. The proposal is deliberately falsifiable. If a simpler stateful governor can preserve equivalent decision quality and prevent the same consequential failures with lower cost, latency, or complexity, the additional Reference-state requirement should not be preferred. Its value would instead be established by showing that substantive pre-transition state exposes or contains material degradation that survives otherwise strong runtime controls.

The objective is therefore not perfect AI or permanent conservatism. It is to preserve a meaningful basis for challenge while the system is still able to change course.

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